

MKUFT Physics-Facing Explanation

Professional reading edition of the evolving public MKUFT canon

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Purpose

This paper explains MKUFT for physicists, technical reviewers, and serious interdisciplinary readers. It focuses on definitions, scope, mathematical status, recovery requirements, and falsifiability.

The foundational quantum-gravity proposal is developed in [Layer Before Law](#). Neither document claims that MKUFT has already derived quantum mechanics, the Born rule, Bell correlations, quantum field theory, general relativity, or a finished theory of quantum gravity.

1. One-line summary

MKUFT proposes as an open research direction that known physical laws may be effective expressions of a deeper constraint-and-relation architecture. The proposal remains incomplete until a lawful mechanism recovers established quantum and gravitational limits and produces a discriminating result beyond existing formalisms.

2. What MKUFT is

MKUFT is a systems-level unification framework. It asks whether spacetime physics, information structure, observer-dependent phenomena, and some boundary regimes can be represented through a deeper typed architecture whose substrate term denotes structured possibility rather than an already detected material medium.

The substrate is not introduced as matter, energy, spacetime, or an extra particle field. It is a candidate pre-physical possibility/constraint layer that may, in a developed model, define admissible configurations and contribute to the emergence of stable effective structures.

This is a research direction, not evidence that such a layer has been physically detected.

3. What MKUFT is not

MKUFT is not a replacement for general relativity or quantum field theory, a completed quantum theory of gravity, consciousness-collapse idealism, simulation theory, a licence for unrestricted causation, a claim that belief overrides physics, or a claim that an architectural category reframing is itself a physical derivation.

Its physical branch must earn scientific status through mathematical definition, accepted-limit recovery, discriminating prediction, and evidence.

4. Four typed layers

The active convention is

$$S \rightarrow I \rightarrow P \rightarrow O.$$

S — Substrate

A useful mathematical representation is the measure space

$$S = (\Omega, \Sigma, \mu),$$

where Ω is a configuration space, Σ a sigma-algebra, and μ a baseline measure or weighting. This representation is not evidence that a separate physical medium exists.

I — Information

Information structures represent patterns, constraints, or relations over the substrate. To keep the layer symbol I distinct from the mathematical function space, write

$$\mathcal{I} = L^2(\Omega, \mu).$$

P — Physical

The physical layer contains measurable events, fields, particles, instruments, and dynamics. Standard physics supplies the baseline predictions in its established domains.

O — Observer

The observer layer contains observer-positioned registration, measurement context, record, attention, interpretation, and bounded state-dependent participation where operationally defined.

A candidate coherence functional is

$$\kappa : \text{States}(\mathcal{H}_O) \rightarrow [0, 1].$$

Observer language cannot replace a missing physical mechanism.

5. Relation to general relativity

General relativity remains the effective description of spacetime curvature and gravitational dynamics in its tested regime.

MKUFT asks whether spacetime geometry itself might be an effective stable configuration generated by deeper relational constraints. That is a research direction, not a derivation. Any developed model must recover Einsteinian or experimentally equivalent gravitational behaviour throughout the established domain.

6. Relation to quantum theory

Quantum theory remains the effective description of amplitudes, probabilistic outcomes, state evolution, correlations, and measurement contexts.

For a state

$$|\psi\rangle = \sum_i c_i |i\rangle,$$

standard quantum theory gives

$$P(i) = |c_i|^2.$$

MKUFT has not derived these relations from its deeper architecture. Any proposed mechanism must recover them rather than merely relabel them.

[Layer Before Law](#) represents an entangled pair as physically distinct P-layer systems with a pair-level I-layer relation. That relation fails as an account of entanglement if it reduces to a locally separable hidden-variable instruction incompatible with Bell tests or permits controllable faster-than-light signalling.

7. Realisation scaffold

For event E , a working unnormalised weight is

$$\tilde{W}(E) = \int D_{\text{phys}}(E | i) W_{SI}(i | S, E) C_O(O | i, E) d\nu(i),$$

with realised probability

$$P_{\text{realized}}(E) = \frac{\tilde{W}(E)}{Z}.$$

Here D_{phys} is the accepted physical contribution conditioned on the information structure, W_{SI} a candidate substrate-to-information weighting, C_O a bounded observer-condition term, and Z the required normalisation.

This is an event-weighting scaffold rather than the complete update mechanism sought by [Layer Before Law](#).

8. Standard-physics limit

If the observer term is event-independent,

$$C_O(O | i, E) = C_0,$$

and the remaining weighting reduces to the accepted physical distribution, then

$$P_{\text{realized}}(E) \approx P_{\text{phys}}(E)$$

within a declared regime, comparison statistic, and tolerance.

Standard-physics recovery is mandatory. For the foundational branch, the proposed unified mechanism must separately recover established quantum and gravitational regimes under their appropriate relational conditions.

9. Observer coupling

Observers are treated as bounded conditions rather than crude forces. Any observer-linked effect must remain lawful, small where a perturbative model is used, and subject to apparatus, environmental, expectancy, and statistical controls.

A candidate linear-response form is

$$C_O(O | i, E) = C_0[1 + \varepsilon \kappa(\rho_O) h(i, E)],$$

where ε is small, κ is a measured observer-state proxy, and h is specified before confirmatory testing.

The architecture alone does not establish that this term is nonzero in nature.

10. Environmental modulation

A candidate environmental damping model is

$$\kappa_{\text{eff}} = \kappa \eta(F), \quad \eta(F) \in [0, 1],$$

where F denotes measured environmental conditions.

If an observer-linked branch is real, preregistered environmental variables may modulate its effect size. Such variables cannot be introduced retrospectively to rescue a null result.

11. Nested micro–macro structure

MKUFT asks whether bounded coherent structures can be represented as local systems nested within wider constraints. Similar formal questions may recur across particles, atoms, cells, organisms, societies, planets, stars, and galaxies.

Recurrence across scales does not establish identity of mechanism. Each physical claim requires its own variables, units, couplings, and evidence.

12. Local closure and boundary emergence

One candidate nesting sequence is:

1. sufficient internal consistency supports local closure;
2. closure stabilises a boundary;
3. the bounded region evolves under both local and wider constraints;
4. the larger system couples to aggregate variables rather than every internal detail.

This is a systems-level architecture. It becomes physical only where a domain-specific model derives and measures the proposed closure and coupling.

13. Anomalous regimes

Any genuine deviation from ordinary behaviour must be sought under tightly defined conditions: measurable gradients, controlled boundary geometry, operationalised observer state where relevant, measured environmental variables, a strong ordinary baseline, and a prespecified effect and falsifier.

Anomaly reports do not validate the substrate, observer coupling, or quantum-gravity branch by themselves.

14. Quantum-gravity research burden

Layer Before Law represents the complete addressed state as

$$U_n = (S_n, I_n, P_n, O_n)$$

and marks the still-unknown update process as

$$U_n \xrightarrow{\text{Update}_{\text{SIPO}}} U_{n+1}.$$

This notation is a placeholder for a missing lawful process, not a completed law.

A developed mechanism must satisfy recovery targets such as

$$\Pi_{PO}[\text{Update}_{\text{SIPO}}(U; R_Q)] \approx Q_{\text{eff}},$$

$$\Pi_{PO}[\text{Update}_{\text{SIPO}}(U; R_G)] \approx G_{\text{eff}},$$

with declared state spaces, dynamics, couplings, comparison norms or statistics, tolerances, and regimes of validity.

The layered entanglement relation must also add more than the standard non-separable joint-state formalism. If no mechanism, derivation, prediction, or improved research route changes, the proposal remains an interpretation or translation rather than new physical theory.

Related public documents

- [Layer Before Law](#)
- [MKUFT Core Extended](#)
- [Mathematical Appendix](#)
- [Experimental Test Programme](#)
- [Typed Traversal and Equation Hygiene](#)
- [Falsification Summary](#)
- [Scientific References and Current Literature](#)